Projection from 3D → 2D Space

Let's say we have some point located at (x, y, z), where $x, y, z \in \mathbb{R}$ and describe a scalars that are applied to the basis vectors $\hat{i}, \hat{j}, \hat{k}$.

We will represent point 'A' as $\begin{bmatrix} x_A \\ y_A \\ z_A \end{bmatrix}$. We need a plane to project this point on, so lets represent the perspective, or camera, 'C' as $\begin{bmatrix} x_C \\ y_C \\ z_C \end{bmatrix}$ with transformation C_T as $\begin{bmatrix} x_{C_T} \\ y_{C_T} \\ z_{C_T} \end{bmatrix}$.

The first step of our projection must obviously be simply to re-orient our perspective in terms of the origin.

$$\Rightarrow A = A - C_T$$

$$\Rightarrow A = \begin{bmatrix} x_A - x_{C_T} \\ y_A - y_{C_T} \\ z_A - z_{C_T} \end{bmatrix}$$

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Now in order to convert our 3D vector to a 2D one, we can apply a matrix transformation. In this case we take a 3×1 matrix and we want the output to be 2×1, hence we can multiply by a 3×2 matrix. We'll call this projection matrix 'P'.

Let $P = \begin{bmatrix} p_{00} & p_{01} & p_{02} \\ p_{10} & p_{11} & p_{12} \end{bmatrix}$

$$\Rightarrow PA = \begin{bmatrix} p_{00} & p_{01} & p_{02} \\ p_{10} & p_{11} & p_{12} \end{bmatrix} \begin{bmatrix} x_A \\ y_A \\ z_A \end{bmatrix}$$

$$\Rightarrow AP = \begin{bmatrix} x_A p_{00} + y_A p_{01} + z_A p_{02} \\ x_A p_{10} + y_A p_{11} + z_A p_{12} \end{bmatrix}$$

So we need to identify this projection matrix. To do this, let's ask: what rotation would take our perspective matrix, C, and convert it to the basis vectors. We can then apply that rotation to our projection matrix.

To apply rotations in 3D, first we need to know how to apply them in 2D.

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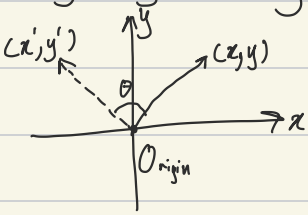
We'll start with the vector $\begin{bmatrix} x \\ y \end{bmatrix}$. We want to rotate it about the origin by some angle, θ. We know how to do this with basic trigonometry



In this case:

$x' = x \times \cos\theta$

$y' = y \times \sin\theta$

So how do we create a rotational matrix out of this?
We need to apply this basic trig to the basis vectors!

In 2D space, we have $\hat{i}$ $\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right)$ and $\hat{j}$ $\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right)$.
To rotate $\hat{i}$, we would need to go from $(1,0)$ to $(\cos\theta, \sin\theta)$
and to rotate $\hat{j}$, we would need to go from $(0,1)$ to $(\cos\theta - 90, \sin\theta - 90)$
and we know $\cos(\theta - 90) = -\sin(\theta)$ and $\sin(\theta - 90) = \cos(\theta)$

$$\Rightarrow \begin{cases} \hat{i}' = \begin{bmatrix} \cos\theta \\ \sin\theta \end{bmatrix} \\ \hat{j}' = \begin{bmatrix} -\sin\theta \\ \cos\theta \end{bmatrix} \end{cases}$$

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To apply this to an arbitrary 2D point, we combine both into a matrix, like so:

Let R_2 be the rotation matrix for a 2D vector,

$$R_2 = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

Now how do we take this and apply it to 3D space? We do the same thing! We apply the rotation one axis at a time. When we do this, we are isolating a plane (for example, to rotate about the x-axis, we isolate the yz plane) and then performing a simple 2D rotation in that plane. (And you can imagine, we can perform rotations in more dimensions by continuing this recursive process.

To isolate an axis, we use the identity matrix and then leave the relevant row and column as is (treating the matrix like a table), and overwrite the rest with the 2D rotational matrix we found.

$$\Rightarrow R_{3_x} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix}, \quad R_{3_y} = \begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix}, \quad R_{3_z} = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

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So to re-orient our perspective, we simply apply all three rotation matrices to our point, A, with respect to our perspective, C.

$$\Rightarrow A' = R_{z_z}(-C_z)\, R_{z_y}(-C_y)\, R_{z_x}(-C_x)\, A$$

And this gives us our projection matrix! Since we have re-oriented our perspective about the origin, we can just ignore the z axis.

$$\Rightarrow P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} R_{z_x}(-C_x)\, R_{z_y}(-C_y)\, R_{z_z}(-C_z)$$

We now have all the ingredients to project our point P on the xy plane, with the origin at C! This basic projection includes no visual depth information, this is called orthographic projection. To get a more realistic perspective projection, we need to divide our rotated point's x and y axis values by the rotated z value.

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